

MIT ACADEMY OF ENGINEERING
FINAL YEAR BTECH SEMESTER - VII EXAMINATION 2026 – 2027
DEPARTMENT OF COMPUTER ENGINEERING
GENERATIVE ARTIFICIAL INTELLIGENCE (PEC Students)
COURSE CODE: 2304422T **NOVEMBER 2026**

DATE:

MAX MARKS: 60

Pedagogy: Project Based Learning

Assessment Type: Project-Based Demonstration, Presentation, and Documentation

1. GENERAL INSTRUCTIONS:

- I. Everybody must come in formal dress and carry a college I-Card and hall ticket for the examination.
- II. Every student must sign on two attendance sheets after the presentation.
- III. Students must design, implement solutions for the decided problem statement and prepare the project finding/summary report/article and presentation slides for the project demonstration.
- IV. Write your Name, Seat Number, PRN number at the top of every page/ file.
- V. Plagiarism is strictly prohibited. If it is found that the contents are copied (both information and program) from internet sources or other students or from any project then assignment will be evaluated directly with zero marks.
- VI. Assessment Panel: The project examination will be conducted jointly by appointed examiners.
- VII. Equal Contribution: Each project member is required to contribute equally in the following aspects: Project Implementation, Demonstration, Presentation and Documentation.
- VIII. Contribution of each student shall be clearly visible.
- IX. Refer to the rubrics for project implementation, demonstration, presentation, documentation and Reviews as mentioned in the given tables.

2. SUBMISSION INSTRUCTIONS:

- I. **Project Code:** must be submitted on Moodle after presentation
- II. **Project Individual Submission** Zip file to be uploaded on Moodle. The zip file must contain a dataset, presentation slides, implementation code (source code and readable .pdf of the code with visible output), project finding/summary report/article and model deployment link, along with project execution steps, on the same day of exam.
- III. All Project reports should be properly designed and as per the rubrics each

and every criteria should be discussed in report.

3. Guidelines for Undertaking Quality Projects (60 Marks)

To ensure that the projects undertaken truly reflect the essence of Generative AI and uphold academic rigor, students are expected to:

- I. Students must form a group of 2–4 members.
- II. Choose a real-world problem aligned with course units and solve it using appropriate Four Generative AI models (LLM, GANs, VAEs, Transformers, Diffusion, GPT, etc.)
- III. Collect a relevant dataset.
- IV. For each technique, students will describe:
 - Architecture & working
 - Selection criteria
 - Mathematical formulation (loss functions, etc.)
 - Implementation outline (pseudo-code/code snippet)
- V. Perform model evaluation using appropriate metrics.
- VI. Provide comparative analysis through tables/graphs.
- VII. Discuss findings, limitations, and ethical considerations.
- VIII. Submit a well-structured report.
- IX. Prepare the following: Code, Dataset, Metrics, PPT, Report, and Ethical Statement.
- X. Submit a ZIP file on Moodle containing all project components.
- XI. Each member must contribute equally to implementation, documentation, and presentation.
- XII. Plagiarism or copying from other groups/internet will lead to zero marks.

4. Student Report Format

1. Cover Page & Abstract
2. Introduction & Problem Definition
3. Dataset Description
4. Selected Generative AI Models
5. Mathematical Foundations
6. Implementation Plan or Pseudocode
7. Evaluation & Results
8. Comparative Metrics Table/Graphs
9. Discussion on Ethical Implications
10. Conclusion & Future Scope
11. References

Evaluation Criteria (60 marks)

Sr. No	Assessment Criteria	Description	Max Marks	COs Mapped	POs Mapped	RBT Level	Detailed Rubric Levels
1	Problem Statement, Objectives, Dataset Selection, Literature Review	Identification of real-world problem or application; clarity and relevance of dataset with proper justification	8	CO1, CO3	PO1, PO2, PO3, PO4, PO5, PO12	L3, L4	• 0–2 (Poor): Problem is unclear or irrelevant; objectives missing; dataset not aligned or insufficient; little/no literature review. • 3–5 (Average/Good): Problem is relevant but lacks clarity; objectives partially clear; dataset is basic with partial justification; literature review has some relevant references, but limited analysis. • 6–8 (Excellent): Problem is well-defined and contextualized; objectives are clear and specific; dataset is relevant, sufficient, and well-justified; literature review is comprehensive, up-to-date, and critically analyzed with gap identification.
2	Model Selection & Description (4 models) (How you apply 4GE AI models in your project))	Correct identification and explanation of GenAI models with reasons for selection	10	CO1	PO1, PO2, PO3, PO5	L3	• 0–3 (Poor): Fewer than 2 models described. Descriptions are vague, incomplete, or irrelevant. No clear justification for applying models in the project. • 4–7 (Average/Good): At least 3–4 models described with basic explanation. Some relevance to the

							project shown, but limited justification. Application of models partially demonstrated (not in depth). • 8–10 (Excellent): All 4 models are clearly described with proper technical depth. Strong justification for why and how each model applies to the project. Comparative insights provided (strengths/limitations). Demonstrates good understanding of applying multiple Generative AI models.
3	System Architecture/ Design, Mathematical Formulation & Integration and clarity	Clear explanation of mathematical components: loss functions, architecture equations	8	CO1, CO2, CO3	PO1, PO2, PO3, PO4, PO5, PO9, PO11, PO12	L3, L4	• 0–2 (Poor): System architecture/design is unclear or missing. Mathematical formulation not attempted or irrelevant. Weak integration between components; lacks clarity in explanation. • 3–5 (Average/Good): Basic architecture presented but lacks detail. Mathematical formulation attempted but partially correct or superficial. Integration shown but limited; clarity moderate. • 6–8 (Excellent): System architecture/design is well-structured, clear, and logically explained. Mathematical formulation is accurate, relevant, and aligned with the model/system. Integration between components is seamless with strong clarity in representation.

							Demonstrates good technical depth and coherence.
4	Implementation Design	Logical design flow, pseudo-code, tools, and model input-output structure	6	CO1, CO2	PO1, PO2, PO3, PO5	L3	• 0–2 (Poor): Implementation design is unclear, incomplete, or missing. No clear link between design and project objectives. Lacks technical details or logical flow. • 3–4 (Average/Good): Basic implementation design presented with some structure. Partial alignment with project objectives. Contains technical elements but limited detail/clarity. • 5–6 (Excellent): Implementation design is well-structured, detailed, and logically explained. Strong alignment with project objectives and chosen models. Demonstrates clarity, technical depth, and feasibility.
5	Evaluation Metrics	Use of relevant and task-appropriate metrics like BLEU, FID, PSNR, etc.	8	CO3	PO2, PO4, PO11	L4	• 0-2 (Poor): Metrics are missing, irrelevant, or incorrectly applied. No justification for why the metrics were chosen. Weak or no interpretation of results. • 3–4 (Average/Good): Some basic evaluation metrics applied (e.g., accuracy, loss, FID, BLEU, etc.). Limited justification for choice of metrics. Results interpreted but with minimal depth or clarity.

							• 5–6 (Excellent): Appropriate and multiple metrics selected for the task (domain-specific, e.g., FID/IS for images, BLEU/ROUGE for text). Strong justification for chosen metrics. Results are clearly analyzed and critically interpreted with insights.
6	Comparative Analysis	Model performance comparison using graphs, tables, and critical interpretation	8	CO1, CO3	PO2, PO4, PO9	L3, L4	• 0–2 (Poor): No comparative study attempted OR comparison is irrelevant/incorrect. Results are presented without analysis. Lacks clarity and justification. • 3–5 (Average/Good): Basic comparison between models/approaches is provided. Some evaluation metrics discussed but limited interpretation. Differences identified but no deep insights. • 6–8 (Excellent): Comprehensive comparative analysis of at least 2–4 models/approaches. Uses appropriate evaluation metrics with proper justification. Clearly highlights strengths, weaknesses, trade-offs of each model. Provides critical insights leading to clear conclusions about model suitability.

7	Deployment with Ethical Concerns	Discussion on bias, misuse, explainability, and impact of GenAI	6	CO2, CO4	PO6, PO7, PO8, PO12	L3, L5	• 0–1 (Poor): Deployment incomplete or unclear. No consideration of ethics, bias, or safety. Lacks clarity in integration or usability. • 2–3 (Average/Good): Deployment attempted and partially functional. Some mention of ethical issues (bias, fairness, privacy) but not deeply addressed. Limited clarity in explaining safeguards. • 4 (Excellent): Deployment is well-executed, integrated, and functional. Ethical concerns clearly addressed (bias mitigation, fairness, safety, responsible AI use). Explanation is clear, with attention to usability and societal impact.
8	Report Quality & Presentation	Structure, formatting, coherence, visual aids, proper referencing	6	CO2, CO3, CO4	PO9, PO10, PO12	L3, L4, L5	• 0–2 (Poor): Report is unstructured or incomplete. Formatting, flow, and language are unclear. Figures/tables/references are missing or inconsistent. Presentation lacks clarity and coherence. • 3–4 (Average/Good): Report has basic structure (problem, methods, results, and conclusion). Formatting is acceptable but minor inconsistencies exist. Some use of visuals, but explanations are limited. Presentation shows partial confidence and clarity.

							5–6 (Excellent): Report is well-structured, polished, and professional. Clear flow with properly formatted sections, visuals, references, and citations. Critical insights and originality are evident. Presentation is confident, clear, engaging, and demonstrates mastery of the subject.
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Reviewer	Designation	Signature / Remarks
Dr.Sunita Barve Dr.Pramod D. Ganjewar	Course Experts	
Pramod Dharmadhiakri (CS)	Academic Autonomy Coordinator	
Dr.Pramod D. Ganjewar	Head of Department	
Ranjana Badre	COE,MIT Academy of Engineering, Alandi,Pune	
Dr.Usha Verma	Head CELT	
Dr.Sunita Barve	Deputy Director A&R, MIT Academy of Engineering, Alandi,Pune	
Dr.Mahesh Goudar	Director, MIT Academy of Engineering,Alandi,Pune	

Prepared by: Dr.Sunita Barve and Mrs. Savita Mane
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